

PHP Backend + AI Usage Assignment

Role

PHP Backend Developer (Laravel / CodeIgniter) with AI-Assisted Development Skills

Objective

Build a simple AI-powered Notes Management System with PHP APIs, AI-generated UI frontend, AI-powered semantic search, and AI-generated note summaries.

Technical Requirements

Backend: Laravel (Preferred) OR CodeIgniter

Database: MySQL/PostgreSQL

Features to Implement

1. Notes CRUD APIs

- Create Note
- Update Note
- Delete Note
- Get Single Note
- Get Notes List

Requirements:

- Validation
- Proper HTTP status codes
- Clean JSON responses
- Pagination support

Semantic Search (AI-Based Search)

Implement AI-powered semantic search using embeddings/vector search or similar techniques.

AI Summary Endpoint

Create endpoint:

POST /api/notes/id/summary

This should generate an AI-based summary of the note content.

Pagination

Implement pagination on notes listing API.

Example: GET /api/notes?page=1&limit=10

AI-Generated Frontend UI

Candidate must create a simple frontend UI using AI tools like Cursor, Copilot, ChatGPT, Claude, Bolt, or v0.

Mandatory Submission Requirements

1. Source Code (GitHub repository preferred)
2. README File including setup instructions, API documentation, database schema, AI tools used, prompts used, and architecture explanation.
3. AI Usage Explanation including where AI was used and how generated code was validated.

Security Expectations

- SQL injection prevention
- API validation
- Rate limiting
- Secure API handling

Bonus Points

- Docker setup
- Unit tests
- Swagger/OpenAPI docs
- Redis usage
- Clean architecture

Time Expectation

Expected completion time: 6–10 hours

Evaluation Criteria

PHP/Laravel Skills – 25%
API Design – 15%
Database Design – 10%
AI Integration – 20%
Semantic Search Quality – 10%

Frontend Quality – 10%
Code Quality & Architecture – 10%

Submission Format

- GitHub repository link
- Setup instructions
- Screenshots/video demo
- README documentation